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In statistics, **sampling techniques** are methods used to select a subset of individuals (a sample) from a larger population to make inferences about the whole group. Sampling is crucial because studying an entire population is often impractical due to time, cost, or logistical constraints.

Sampling techniques are broadly classified into two categories:

1. Probability Sampling (Random Sampling)

Every member of the population has a **known, non-zero chance** of being selected. These methods are preferred for statistical analysis because they minimize **bias**.

Types of Probability Sampling:

✅ **Simple Random Sampling**

- Every individual has an **equal chance** of being selected.
- Example: Drawing names from a hat, using random number generators.

✅ **Stratified Sampling**

- The population is divided into **subgroups (strata)** based on shared characteristics (e.g., age, gender).
- Samples are then randomly selected from each stratum.
- Ensures representation of all key groups.

✅ **Cluster Sampling**

- The population is divided into **clusters** (e.g., schools, cities).
- A few clusters are randomly selected, and **all individuals** within them are studied.
- Useful when populations are geographically spread out.

✅ **Systematic Sampling**

- Selecting every **k-th element** from a list (e.g., every 10th customer).
- Example: Surveying every 5th student in a school register.

✅ **Multistage Sampling**

- Combines two or more sampling methods (e.g., first cluster sampling, then random sampling within clusters).
- Used in large-scale surveys like national censuses.

In statistics, **SRSWOR (Simple Random Sampling Without Replacement)** and **SRSWR (Simple Random Sampling With Replacement)** are two fundamental probability sampling techniques under **Simple Random Sampling**. Both involve random selection, but they differ in whether selected units are **returned to the population** for possible reselection.

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1. SRSWOR (Simple Random Sampling Without Replacement)

✅ **Definition:**

- Each unit (individual/item) is selected **only once** and **not returned** to the population.
- Once chosen, it cannot be picked again.

✅ **Key Features:**

- **No duplicates** in the sample.
- Sample size (**n**) must be ≤ Population size (**N**).
- Reduces variability compared to SRSWR.

✅ **Example:**

- **Election Polling:** Selecting 100 voters from 1000 **without repeating any voter**.

✅ **Probability of Selection:**

- For each draw, the probability changes because the pool shrinks.
- **First draw:** P=1 N=1
- **Second draw:** P=1N−1$P=N-1$1
- **n-th draw:** P=1N−n+1$P=N-n+1$1

✅ **Advantages:**

- More efficient (lower variance) than SRSWR.
- Avoids duplication, better for practical surveys.

✅ **Disadvantages:**

- Requires a **fixed population list**.
- Not suitable if population size is unknown.

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2. SRSWR (Simple Random Sampling With Replacement)

✅ **Definition:**

- Each unit is selected **independently**, and **returned to the population** before the next draw.
- The same unit can appear **multiple times** in the sample.

✅ **Key Features:**

- **Possible duplicates** in the sample.
- Sample size (**n**) can be **larger than population size (N)**.
- Used in bootstrapping and simulations.

✅ **Example:**

- **Quality Control:** Testing 50 light bulbs from a batch of 100, where each bulb is tested and put back.

✅ **Probability of Selection:**

- Each draw is independent, so probability remains constant.
- For any single draw: P=N=1 (same for all selections).

✅ **Advantages:**

- Simple to implement (no need to track previous selections).
- Useful when population size is dynamic.

✅ **Disadvantages:**

- Higher sampling variance (less efficient than SRSWOR).
- Duplicates can lead to redundant data.

Comparison Table: SRSWOR vs. SRSWR

Feature	SRSWOR	SRSWR
Replacement?	✗ No (once selected, excluded)	✓ Yes (can be selected again)
Sample Size (n)	Must be ≤ Population (N)	Can be > N
Duplicates?	No	Yes
Variance	Lower (more precise)	Higher
Probability per draw	Changes (depends on remaining pool)	Constant (1N*N*1)
Common Use Cases	Surveys, polls, experiments	Bootstrapping, simulations

When to Use Which?

- **Use SRSWOR** when:
  - You need **unique samples** (e.g., human surveys).
  - Population is **fixed and known**.
  - Higher precision is required.
- **Use SRSWR** when:
  - Population is **unlimited or dynamic**.
  - You need **resampling methods** (e.g., bootstrapping in ML).
  - Simplifying selection process (e.g., lottery systems).

Sampling Error in statistics

In statistics, **sampling error** refers to the difference between a **sample statistic** (e.g., sample mean, sample proportion) and the **true population parameter** it estimates, due to the fact that the sample is only a subset of the population.

Key Points:

1. **Caused by Random Sampling**:
   - Sampling error occurs because we use a sample (a smaller group) to represent the entire population.
   - Even with a well-chosen random sample, there will always be some variability between the sample and the true population value.
2. **Not Due to Mistakes**:
   - Unlike **non-sampling errors** (e.g., measurement errors, biased sampling), sampling error is a natural and unavoidable part of statistical inference.

3. **Decreases with Larger Samples**:
    - As the sample size increases, the sampling error generally decreases (due to the **Law of Large Numbers**).
    - However, the relationship is not linear—increasing sample size reduces error at a decreasing rate.

Formula:

For a sample mean ($\bar{x}$) estimating the population mean (μ):

Sampling Error=$\bar{x}-\mu$

Example:

Suppose the true average height of all adults in a city is **170 cm** ($\mu=170$). If you take a random sample of 100 people and find their average height is **168 cm** ($\bar{x}=168$), the sampling error is:

168−170=−2 cm168−170=−2 cm

Reducing Sampling Error:

- Increase the **sample size** (n).
- Use **random sampling** to avoid bias.
- Apply **stratified sampling** (if subgroups exist) to ensure better representation.

Sampling Error vs. Standard Error:

- **Sampling Error**: The actual difference between the sample and population.
- **Standard Error**: A measure of the variability of sampling error across different samples (estimated using the sample standard deviation and size).

Why It Matters:

Understanding sampling error helps quantify uncertainty in estimates, which is crucial for constructing **confidence intervals** and conducting **hypothesis tests**.

Cochran's Minimum Sample Size Formula

Cochran's formula is used to determine the **minimum sample size** required for a survey or experiment when sampling from a **large population**, ensuring results are statistically significant. It is widely used in **market research, polling, and social sciences**.

Cochran's Formula (For Large Populations)

The basic formula for calculating the minimum sample size (n0*n*0) is:

$$n_0 = \frac{Z^2 \cdot p \cdot (1-p)}{e^2}$$

Where:

- n_o = Minimum required sample size
- Z = **Z-score** (critical value for desired confidence level, e.g., 1.96 for 95% confidence)
- p = **Estimated proportion** of the population (if unknown, use p=0.5 for maximum variability)
- e = **Margin of error** (acceptable error percentage, e.g., 0.05 for ±5%)

Adjusted Formula for Finite Populations

If the population (N*N*) is **small or known**, Cochran's formula is adjusted to:

$$n = \frac{n_0}{1 + \frac{(n_0 - 1)}{N}}$$

Where:

- n = Adjusted sample size
- N = Total population size
- n0 = Initial sample size from the first formula

Steps to Apply Cochran's Formula

1. **Choose Confidence Level (e.g., 95%)** → Get **Z-score** (1.96 for 95%).
2. **Estimate *p*** (if unknown, use p=0.5, which gives the largest sample size).
3. **Set Margin of Error (e*e*)** (e.g., 0.05 for 5% error).
4. **Plug values into Cochran's formula** to get n0*n*0.
5. **If population is finite**, adjust using the second formula.

Example Calculation

Scenario: You want to survey customer satisfaction in a company with **10,000 customers**. You need **95% confidence** and a **5% margin of error**.

1. **Z-score (95% confidence)** = 1.96
2. **Assume p=0.5** (maximum variability)
3. **Margin of error (e)** = 0.05

Step 1: Calculate initial sample size (n0):

n0=(1.96)2×0.5×(1−0.5)(0.05)2=3.8416×0.250.0025=384.16≈∗∗385∗∗*n*0=(0.05)2(1.96)2×0.5×(1−0.5) =0.00253.8416×0.25=384.16≈∗∗385∗∗

Step 2: Adjust for finite population (N=10,000):

n=3851+(385−1)10,000=3851+0.0384≈**371**n=1+10,000(385−1)385=1+0.0384385≈**371**

Final Required Sample Size = 371 customers

Key Takeaways

✅ **Use Cochran's formula when:**

- The population is **large or unknown** (use first formula).
- The population is **small and known** (use adjusted formula).

✅ **If p is unknown**, assume p=0.5 (most conservative estimate).

✅ **Higher confidence (Z↑) or lower margin of error (e↓)** increases sample size.

✅ **For small populations**, the adjusted sample size reduces the requirement.

When Not to Use Cochran's Formula?

❌ **Stratified or clustered sampling** (use other sampling techniques).
❌ **Very small populations** (use census instead).
❌ **Non-probability sampling** (e.g., convenience sampling).

Methods of Collecting Data in Research

Data collection is a crucial step in research, business, and decision-making. The choice of method depends on the **research objectives, resources, and type of data needed**.

Data collection methods are broadly classified into **two categories**:

1. **Primary Data Collection** (First-hand data)
2. **Secondary Data Collection** (Pre-existing data)

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1. Primary Data Collection Methods

Data collected **directly from original sources** for a specific study.

A. Surveys & Questionnaires

- **Description:** Structured questions (online, phone, paper).
- **Types:**
    - **Structured (Closed-ended):** Multiple-choice, Likert scales.
    - **Unstructured (Open-ended):** Free-text responses.
- **Example:** Customer feedback forms, election polls.

B. Interviews

- **Description:** One-on-one or group discussions.
- **Types:**
  - **Structured:** Fixed questions.
  - **Semi-structured:** Flexible follow-ups.
  - **Unstructured:** Free-flowing (e.g., psychological studies).
- **Example:** Job interviews, market research.

C. Observations

- **Description:** Recording behavior **without interference**.
- **Types:**
  - **Participant Observation:** Researcher engages (e.g., ethnographic studies).
  - **Non-Participant Observation:** Detached recording (e.g., CCTV analysis).
- **Example:** Retail store customer behavior tracking.

D. Experiments

- **Description:** Controlled tests to establish **cause-effect relationships**.
- **Types:**
  - **Lab Experiments:** Strictly controlled (e.g., drug trials).
  - **Field Experiments:** Real-world settings (e.g., A/B testing in marketing).
- **Example:** Testing a new fertilizer's effect on crop yield.

E. Focus Groups

- **Description:** Small group discussions (6–10 people) led by a moderator.
- **Example:** Testing product concepts before launch.

F. Case Studies

- **Description:** In-depth analysis of a **single unit** (person, company, event).
- **Example:** Studying Tesla's business model.

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2. Secondary Data Collection Methods

Using **pre-existing data** collected by others.

A. Published Sources

- **Books, Journals, Reports** (e.g., World Bank reports).
- **Government Publications** (e.g., Census data).

B. Online Databases

- **Google Scholar, Statista, Kaggle.**
- **Company Records** (e.g., annual reports).

C. Media Sources

- **News articles, documentaries.**

D. Public & Private Records

- **Hospital records, school registries.**

Comparison: Primary vs. Secondary Data

Factor	Primary Data	Secondary Data
Source	Collected firsthand	Already available
Cost	Expensive (time, effort)	Low-cost
Time	Lengthy	Quick access
Accuracy	High (customized)	May be outdated or biased
Purpose	Specific research needs	Broad reference

How to Choose the Right Method?

✓ **For original insights** → Primary data (surveys, experiments).
✓ **For quick, low-cost data** → Secondary data (reports, databases).
✓ **For behavioral studies** → Observations, focus groups.
✓ **For statistical analysis** → Surveys, existing datasets.

Z-Value (Standard Value) in Statistics

The **Z-value** (or **Z-score**) measures how many **standard deviations** a data point is from the mean of a distribution. It is a key concept in **hypothesis testing, confidence intervals, and normal distribution analysis**.

Formula for Z-Value

The Z-score is calculated as:

$$Z = \frac{X - \mu}{\sigma}$$

- **X** = Individual data point
- **μ** = Mean of the population
- **σ** = Standard deviation of the population

For sample data, use:

$$Z = \frac{X - \bar{X}}{s}$$

where $\bar{X}$ = sample mean, s = sample standard deviation.

Interpretation of Z-Value

Z-Score	Meaning	Example
Z = 0	Data point = Mean	Exactly average performance
Z = +1.0	1 standard deviation **above** mean	Better than ~84% of the data
Z = -1.0	1 standard deviation **below** mean	Worse than ~84% of the data
Z = +2.0	2 standard deviations **above** mean	Top ~2.5% of the data
Z = -2.0	2 standard deviations **below** mean	Bottom ~2.5% of the data

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Uses of Z-Value

✅ **Standardization:** Compare data from different scales (e.g., SAT vs. ACT scores).
✅ **Outlier Detection:** Values with $|Z|>3$ are extreme outliers.
✅ **Hypothesis Testing:**

- Used in **Z-tests** (when population parameters are known).
- Determines **p-values** for significance testing.
  ✅ **Confidence Intervals:**
- **95% CI** uses Z=1.96.
- **99% CI** uses Z=2.576.

Example Calculation

Problem: A class has a mean score (μ) of **70** and a standard deviation (σ) of **10**. What is the Z-score for a student who scored **85**?

Solution:

$$Z = \frac{85 - 70}{10} = \frac{15}{10} = **1.5**$$

Interpretation: The student scored **1.5 standard deviations above the mean**.

Z-Value vs. T-Value

Feature	Z-Value	T-Value
When Used	Large samples (n≥30) or known σ	Small samples (n<30) or unknown σ
Distribution	Standard Normal Distribution (Z)	Student's T-Distribution (t)
Critical Values	Fixed (e.g., 1.96 for 95% CI)	Depends on degrees of freedom (dof)

Key Takeaways

✓ A **Z-score** tells how unusual or typical a value is in a dataset.
✓ **Positive Z** = Above mean | **Negative Z** = Below mean.
✓ Used in **quality control, finance (risk modeling), and machine learning (feature scaling)**.

Data Distribution plots in statistics

In statistics, **data distribution plots** are graphical representations that show how data points are spread or distributed across different values. These plots help visualize key characteristics of a dataset, such as:

- **Central tendency** (mean, median, mode)
- **Spread** (range, variance, outliers)
- **Shape** (symmetry, skewness, modality)

Common Types of Distribution Plots:

1. **Histogram**

- **Purpose**: Shows the frequency of data points in predefined bins (intervals).
- **Use Case**: Identifying the shape of the distribution (e.g., normal, skewed).
- **Example**:

python

```
import matplotlib.pyplot as plt

data = [1, 1, 2, 3, 3, 3, 4, 5]

plt.hist(data, bins=5)

plt.show()
```

2. **Box Plot (Box-and-Whisker Plot)**

- **Purpose**: Displays the median, quartiles, and outliers of a dataset.
- **Use Case**: Comparing distributions across groups or detecting outliers.
- **Components**:
    - **Box**: Represents the interquartile range (IQR; 25th–75th percentile).

- **Whiskers**: Extend to the smallest/largest non-outlier values.
- **Outliers**: Points beyond 1.5×IQR.

3. **Density Plot (Kernel Density Estimation, KDE)**

- **Purpose**: Smooths a histogram to estimate the probability density function.
- **Use Case**: Visualizing the underlying distribution shape.

4. **Q-Q Plot (Quantile-Quantile Plot)**

- **Purpose**: Compares sample quantiles to a theoretical distribution (e.g., normal).
- **Use Case**: Checking if data follows a specific distribution.
- **Interpretation**: Points on a straight line suggest a good fit.

5. **Violin Plot**

- **Purpose**: Combines a box plot with a KDE to show distribution density.
- **Use Case**: Comparing distributions with richer detail than a box plot.

Key Terms Linked to Distribution Plots:

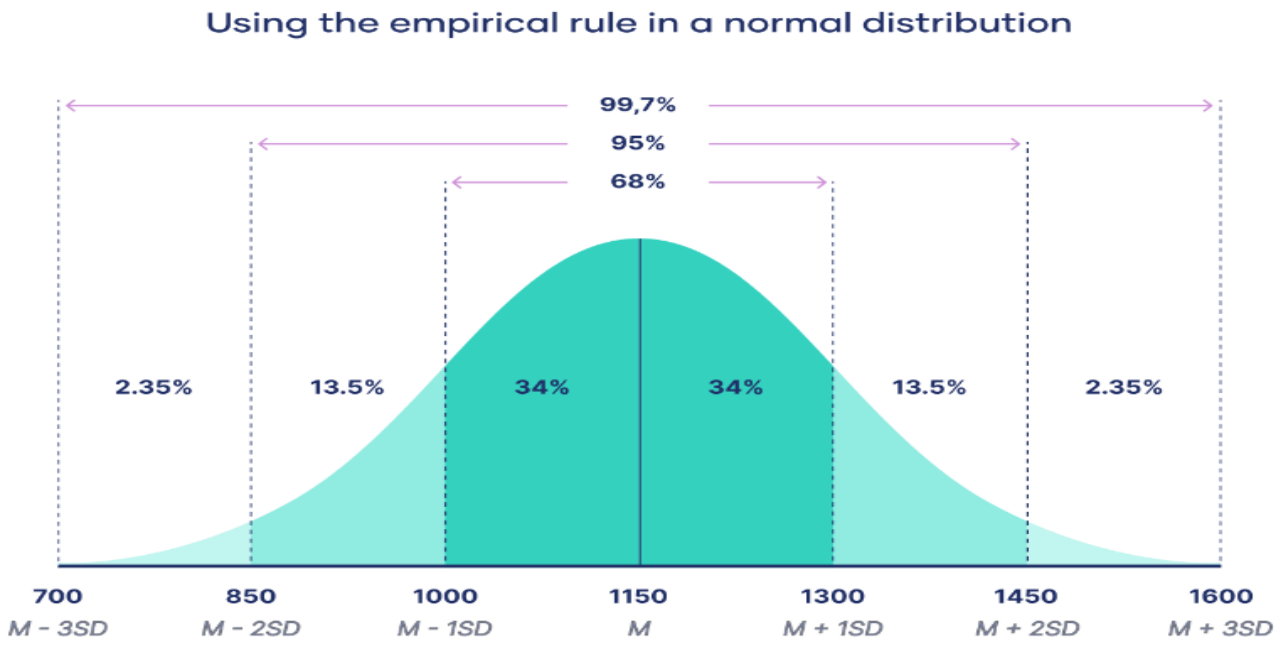
- **Skewness**: Asymmetry in the distribution (left-skewed vs. right-skewed).
- **Kurtosis**: "Tailedness" (peakedness) of the distribution.
- **Modality**: Number of peaks (unimodal, bimodal, multimodal).

Example Workflow:

1. **Explore Data**: Start with a histogram/KDE to see the shape.
2. **Check Normality**: Use a Q-Q plot.
3. **Compare Groups**: Use box plots or violin plots.
4. **Identify Outliers**: Box plots are especially useful.

Normal Distribution & Properties

- **Definition**: The normal distribution is a continuous probability distribution that is symmetrical around its mean, with data near the mean more frequent in occurrence than data far from the mean.
- **Properties**:
  - Mean = Median = Mode
  - The area under the curve is 1.
  - About 68% of the data falls within one standard deviation of the mean, 95% within two, and 99.7% within three.



- **Example**: Heights of people often follow a normal distribution.

The normal distribution, also known as the Gaussian distribution, is a fundamental concept in statistics and probability theory. It describes how the values of a variable are distributed and is characterized by its symmetric, bell-shaped curve. Here's a detailed look at its key features:

Characteristics of the Normal Distribution

1. **Symmetry**: The normal distribution is perfectly symmetric around its mean. This means that the left and right sides of the distribution are mirror images of each other.
2. **Mean, Median, and Mode**: In a normal distribution, the mean (average), median (middle value), and mode (most frequent value) are all equal and located at the center of the distribution.
3. **Bell-Shaped Curve**: The graph of a normal distribution is a bell-shaped curve. It rises smoothly to a peak at the mean and then falls off symmetrically.
4. **Standard Deviation**: The spread of the distribution is determined by the standard deviation (σ). A larger standard deviation results in a wider, flatter curve, while a smaller standard deviation results in a steeper, narrower curve. The standard deviation measures the average distance of each data point from the mean.
5. **Probability Density Function (PDF)**: The PDF of the normal distribution is given by the formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

   Where, μ is the mean, σ is the standard deviation, and e is the base of the natural logarithm.
6. **Empirical Rule**: it provides a quick way to understand the spread of data in a normal distribution.

Importance and Applications

- **Statistical Inference**: Many statistical tests and methods, such as hypothesis testing and confidence intervals, assume that data follows a normal distribution.
- **Natural Phenomena**: Many natural and human-made phenomena are approximately normally distributed, such as heights, test scores, and measurement errors.
- **Central Limit Theorem**: This theorem states that the distribution of the sample mean will approach a normal distribution as the sample size increases, regardless of the original distribution of the data.

Empirical Rule and Outliers

1. Empirical Rule (68-95-99.7 Rule)

The **Empirical Rule** applies to **normally distributed** (bell-shaped) data and states:

- **68%** of data falls within **±1 standard deviation (σ)** of the mean (μ).
- **95%** of data falls within **±2σ** of the mean.
- **99.7%** of data falls within **±3σ** of the mean.

Example:
If a dataset has:

- Mean (μ) = 50
- Standard deviation (σ) = 10

Then:

- **68%** of values are between **40 and 60** (μ ± σ).
- **95%** are between **30 and 70** (μ ± 2σ).
- **99.7%** are between **20 and 80** (μ ± 3σ).

2. Identifying Outliers Using the Empirical Rule

Since **99.7%** of data should be within **±3σ**, any data point outside this range is considered a **potential outlier**.

Example:
Using the same mean (μ = 50) and σ = 10:

- **Lower Bound:** μ − 3σ = 50 − 30 = **20**
- **Upper Bound:** μ + 3σ = 50 + 30 = **80**

Any value **below 20 or above 80** is a possible outlier.

3. Alternative Methods for Outlier Detection

If data is **not normally distributed**, other methods can be used:

- **IQR (Interquartile Range) Method**:
  - Outliers are values **below Q1 − 1.5×IQR** or **above Q3 + 1.5×IQR**.
- **Z-Score Method**:
  - If |Z-score| > **3**, the point is an outlier.

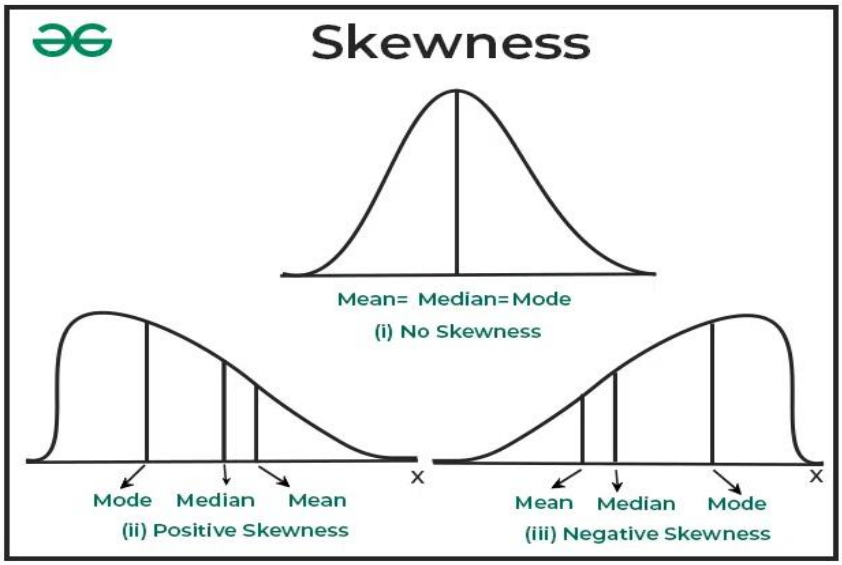
Key Takeaways

- The Empirical Rule works **only for normal distributions**.
- Outliers are extreme values that deviate significantly from the pattern.
- For non-normal data, use **IQR or Z-scores** instead.

Skewness and Kurtosis

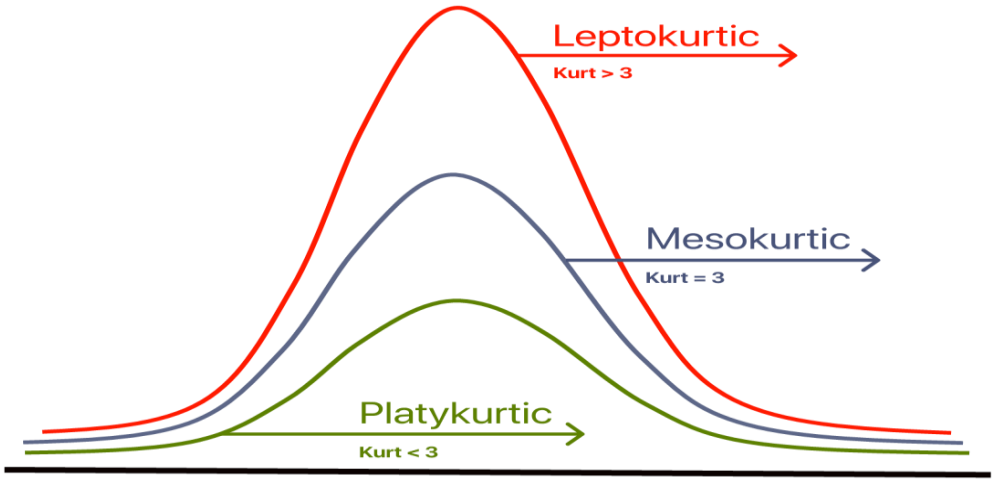
a. Skewness

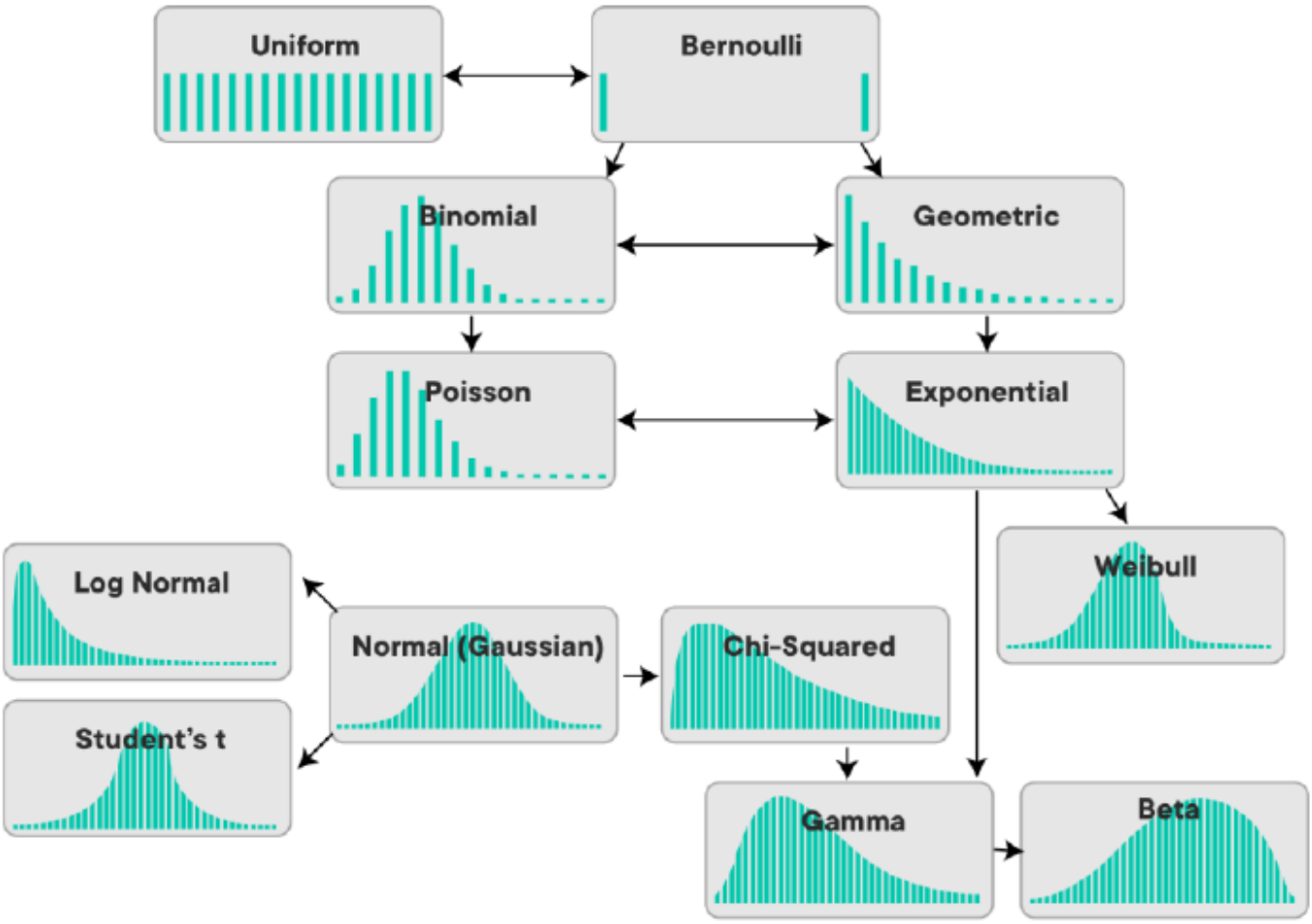
- **Definition**: Measures the asymmetry of the data distribution.
  - **Positive Skew**: Long tail on the right.
  - **Negative Skew**: Long tail on the left.
  - **Symmetric**: Data is evenly distributed.



b. Kurtosis

- **Definition**: Measures the "tailedness" of the data distribution.
  - **Leptokurtic** (K>3): More outliers, sharper peak.
  - **Mesokurtic** (K=3): Normal distribution.
  - **Platykurtic** (K<3): Fewer outliers, flatter peak





Central Limit Theorem (CLT)

It states that:

"If you take sufficiently large random samples from any population (regardless of its original distribution), the distribution of the sample means will approximate a normal distribution (bell curve), with the mean equal to the population mean and the standard deviation decreasing as sample size increases."

Key Points:

1. **Applies to Any Population Distribution**
    - Whether the original data is skewed, uniform, binomial, or even weirdly shaped, the **sample means** will tend toward normality as sample size grows.
2. **Requires Large Enough Sample Size**
    - Typically, **n ≥ 30** is considered sufficient for CLT to hold.
    - For highly skewed distributions, larger samples may be needed.
3. **Mean of Sample Means = Population Mean (μ)**
    - If you take many samples and average their means, it will converge to the true population mean.
4. **Standard Deviation of Sample Means (Standard Error)**
    - The spread of sample means is given by:

Standard Error (SE)= σ n

where:

- σ = population standard deviation
- n = sample size

Why is CLT Important?

- **Makes Inferences Possible:** Even if we don't know the population distribution, we can use normal distribution properties for hypothesis testing (e.g., t-tests, confidence intervals).
- **Justifies "Averages" in Real Life:** Many real-world processes (e.g., quality control, election polling) rely on averages behaving normally.

Example of CLT in Action

Scenario: Suppose we measure the height of 1000 people (population). The distribution is **right-skewed** (few very tall individuals).

- **Step 1:** Take **many random samples** (e.g., 200 samples) of size **n = 40**.
- **Step 2:** Calculate the **mean height** for each sample.
- **Result:** The distribution of those **sample means** will be ~normal, centered at the true average height!

CLT vs. Law of Large Numbers

- **CLT:** Describes the **shape** (normal distribution) of sample means.
- **Law of Large Numbers:** Says that as **n** → ∞, the sample mean converges to the population mean (but doesn't specify the distribution shape).

When Does CLT Fail?

- **Small sample sizes (n < 30)** – Normality isn't guaranteed.
- **Heavy-tailed distributions (e.g., Cauchy distribution)** – CLT may not apply even for large n.

Key Takeaway

The CLT allows statisticians to make predictions and inferences about populations **even when the original data isn't normal**, as long as sample sizes are large enough.

Normality Testing

Normality testing checks whether a dataset follows a **normal (Gaussian) distribution**, which is a key assumption for many statistical tests (e.g., t-tests, ANOVA, regression).

---

1. Why Test for Normality?

Many statistical methods assume normality because:

✅ **Parametric tests** (like t-tests) require normally distributed data.

✅ Ensures **reliable confidence intervals & p-values**.

✅ Helps detect **skewness, outliers, or unusual patterns**.

If data is **not normal**, you may need:

- Non-parametric tests (e.g., Mann-Whitney U, Kruskal-Wallis).
- Data transformations (e.g., log, square root).

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2. How to Test for Normality?

A. Graphical Methods (Visual Checks)

(1) Histogram + Overlaid Normal Curve

- **Check:** Does the histogram match the bell curve?
- **Example:**
  - Normal data → Symmetric, peak at mean.
  - Skewed data → Tails longer on one side.

(2) Q-Q Plot (Quantile-Quantile Plot)

- **Check:** Do data points follow the **straight diagonal line**?
  - If **points deviate sharply**, data is non-normal.
  - **Example:**
    - Normal data → Points align with the line.
    - Right-skewed → Points curve upward.

B. Statistical Tests (Formal Tests)

These give **p-values** to decide normality.

Test	What it Checks	When to Use	Interpretation
Shapiro-Wilk	Overall fit to normality	Small samples (<50)	**$p < 0.05$ → Not normal**
Kolmogorov-Smirnov (K-S)	Compares to a normal distribution	Large samples	Sensitive to tails
Anderson-Darling	Emphasizes tail behavior	When tails matter	Stricter than Shapiro-Wilk

D'Agostino-Pearson	Tests skewness & kurtosis	General use	Combines skew + kurtosis tests

Rule of Thumb:

- **p > 0.05** → Fail to reject normality (data is ~normal).
- **p ≤ 0.05** → Reject normality (data is **not normal**).

---

3. What If Data is Not Normal?

Option 1: Transform the Data

- **Log transform** (for right-skewed data).
- **Square root** (for count data).
- **Box-Cox transformation** (general-purpose).

Option 2: Use Non-Parametric Tests

- **Mann-Whitney U test** (instead of t-test).
- **Kruskal-Wallis** (instead of ANOVA).
- **Spearman's rank** (instead of Pearson correlation).

Option 3: Increase Sample Size

- **CLT applies** → Means become normal with large n (~30+).

5. Key Takeaways

- ✓ **Normality matters** for parametric tests.
  ✓ **Check visually** (histogram, Q-Q plot) and **statistically** (Shapiro-Wilk).
  ✓ If **non-normal**, transform data or use non-parametric tests.

Measures of Distance: Euclidean, Manhattan, and Minkowski Distance

Distance measures are fundamental in machine learning, data mining, and statistics for quantifying the similarity or dissimilarity between data points. Three commonly used distance metrics are:

1. **Euclidean Distance**
2. **Manhattan Distance**
3. **Minkowski Distance**

Let's explore each in detail.

---

1. Euclidean Distance (L_2 Norm)

- The most common distance measure, representing the shortest straight-line distance between two points in Euclidean space.
- Formula for two points P = (p1,p2,...,pn) and Q = (q1,q2,...,qn):

$$D_{\text{Euclidean}}(P,Q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$$

- **Example (2D space):**
  P = (1,3), Q = (4,7)

  D = (1−4) ^2 + (3−7) ^2 = sqrt (9+16) = 5

Use Cases:

  - K-Nearest Neighbors (KNN)
  - Clustering (K-Means)
  - When features are continuous and on similar scales.

---

2. Manhattan Distance (L_1 Norm, Taxicab Distance)

- Measures distance as the sum of absolute differences along each axis (like moving in a grid-like path).
- Formula:

$$D_{\text{Manhattan}}(P,Q) = \sum_{i=1}^{n}|p_i - q_i|$$

- **Example (2D space):**
  P = (1,3), Q = (4,7)

  D = |1−4| + |3−7| = 3+4 = 7
- **Use Cases:**
  - Robust to outliers (less sensitive than Euclidean).
  - Useful in high-dimensional spaces (e.g., text mining).
  - Works well with grid-like movement (e.g., urban navigation).

---

3. Minkowski Distance (Generalized Form)

- A generalization of Euclidean and Manhattan distances using a parameter p.
- Formula:

$$D_{\text{Minkowski}}(P,Q) = \left(\sum_{i=1}^{n} |p_i - q_i|^p\right)^{1/p}$$

- **Special Cases:** In above formula if,
  - p=1: **Manhattan Distance**
  - p=2: **Euclidean Distance**
  - p→∞: **Chebyshev Distance** (max absolute difference along any axis)
- **Use Cases:**
  - Flexibility in choosing p based on data characteristics.
  - Used in algorithms where distance sensitivity needs tuning.

Comparison Summary

Distance Metric	When to Use
Euclidean (L_2)	Natural straight-line distances.
Manhattan (L_1)	Grid-like paths, robust to outliers.
Minkowski (L_p)	Grid-like paths, robust to outliers.

Key Takeaways

- **Euclidean** is best for "as-the-crow-flies" distances.
- **Manhattan** is better for high-dimensional or noisy data.
- **Minkowski** provides a flexible framework for different norms.

What is Covariance?

Covariance is a statistical measure that quantifies the degree to which two random variables change together. It indicates:

- **Direction of relationship:** Whether the variables tend to increase or decrease simultaneously.
- **Strength of linear association:** How strongly the variables are related in a linear fashion.

However, covariance does **not** measure the strength of the relationship on a standardized scale (unlike correlation).

Covariance Formula

For two random variables X and Y with n samples:

$$\text{Cov}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

where:

- $\bar{X}$ and $\bar{Y}$ are the means of X and Y.

Sample Covariance (unbiased estimator, used in practice):

$$\text{Cov}_{\text{sample}}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

Interpretation of Covariance

Covariance Value	Interpretation
Positive	X and Y tend to move in the **same direction** (if X increases, Y tends to increase).
Negative	X and Y tend to move in **opposite directions** (if X increases, Y tends to decrease).
Zero	No linear relationship (but they may have a nonlinear relationship).

Example Calculation

Let's compute covariance for:

X = [1,2,3,4,5], Y = [5,4,3,2,1]

1. **Compute means:**

$$\bar{X} = \frac{1+2+3+4+5}{5} = 3, \quad \bar{Y} = \frac{5+4+3+2+1}{5} = 3$$

2. **Compute deviations and multiply:**

(1−3) (5−3) = (−2) (2) = −4

(2−3) (4−3) = (−1) (1) = −1

(3−3) (3−3) = (0) (0) = 0

(4−3) (2−3) = (1) (−1) = −1

(5−3) (1−3) = (2) (−2) = −4

3. **Sum and average:**

$$\text{Cov}(X,Y) = \frac{-4-1+0-1-4}{5} = \frac{-10}{5} = -2$$

Interpretation: Strong **negative** linear relationship.

Covariance vs. Correlation

Covariance	Correlation
Measures **direction** and **linear relationship**.	Measures **direction** and **strength** (normalized between -1 and 1).
Depends on **units** of variables.	Unitless (standardized).
Range: $(-\infty, +\infty)$.	Range: $[-1, 1]$.

Correlation Formula:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where σX, σY are standard deviations.

Applications of Covariance

1. **Portfolio Theory (Finance):** Measures how assets move together.
2. **Principal Component Analysis (PCA):** Used in dimensionality reduction.
3. **Machine Learning:** Feature selection and understanding relationships in data.

Key Takeaways

- Covariance indicates **direction** but not the **strength** of a linear relationship.
- **Positive covariance** → Variables move together.
- **Negative covariance** → Variables move oppositely.
- **Zero covariance** → No linear relationship (but check for nonlinear patterns).

Hypothesis Testing

Hypothesis testing is a statistical method used to **make decisions** about a population based on sample data. It helps determine whether observed patterns are **real (statistically significant)** or just due to random chance.

1. Key Concepts

A. Null Hypothesis (H_0)

- The default assumption (e.g., "no effect," "no difference").
- Example:
  - *"The new drug has no effect on blood pressure."*
  - *"Men and women earn equal salaries."*

B. Alternative Hypothesis (H_1 or H_a)

- The claim we want to test (e.g., "there is an effect").
- Example:
  - *"The new drug reduces blood pressure."*
  - *"Men earn more than women."*

C. Significance Level (α)

- The threshold for rejecting H_0 (usually $\alpha = 0.05$ or 5%).
- If **p-value < α**, we reject H_0.

D. p-value

- Probability of observing the data (or more extreme) **if H_0 were true**.
- **Low p-value ($\leq \alpha$)** → Evidence against H_0.
- **High p-value ($> \alpha$)** → Fail to reject H_0.

---

2. Steps in Hypothesis Testing

1. **State the Hypotheses**
   - Define H_0 and H_1.
   - Example:
     - H_0: $\mu = 50$ (population mean is 50).
     - H_1: $\mu \neq 50$ (two-tailed test).
2. **Choose Significance Level (α)**
   - Common: $\alpha = 0.05$ (5% risk of false rejection).
3. **Select the Right Test**
   - **Z-test** (known population σ, large n).
   - **t-test** (unknown σ, small n).
   - **Chi-square** (categorical data).
   - **ANOVA** (comparing >2 groups).

4. **Calculate Test Statistic**
    - Example (t-test):

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

        - $\bar{X}$ = sample mean
        - μ = hypothesized mean
        - s = sample standard deviation
        - n = no. of observations in a sample
5. **Find p-value & Make Decision**
    - If **p ≤ α → Reject H_0** (significant result).
    - If **p > α → Fail to reject H_0**.
6. **Interpret the Result**
    - Example:
        - *"We reject H_0 (p = 0.03), concluding the drug lowers blood pressure."*

Critical Region (Rejection Region) in Hypothesis Testing

The **critical region** (or **rejection region**) is the range of values of a test statistic (e.g., *z*, *t*, F, χ^2) for which we **reject the null hypothesis (H0)**. It is determined by:

1. The **significance level (α)** (e.g., 0.05, 0.01).
2. The **type of test** (left-tailed, right-tailed, or two-tailed).

---

1. How to Find the Critical Region?

Step 1: Choose the Significance Level (α)

- Common choices: **α = 0.05 (5%)**, **α = 0.01 (1%)**
- This is the probability of **wrongly rejecting H0** (Type I error).

Step 2: Determine the Type of Test

Test Type	Alternative Hypothesis (Ha)	Critical Region Location
Left-tailed	Ha:μ<μ0	Extreme **left** tail
Right-tailed	Ha:μ>μ0	Extreme **right** tail
Two-tailed	Ha: μ ≠ μ0	**Both** tails (α/2 in each)

Step 3: Find the Critical Value(s)

- Use statistical tables (z-table, t-table) or software (Excel, Python, R).
- The critical value separates the **acceptance region** from the **rejection region**.

Example (Z-Test, α = 0.05)

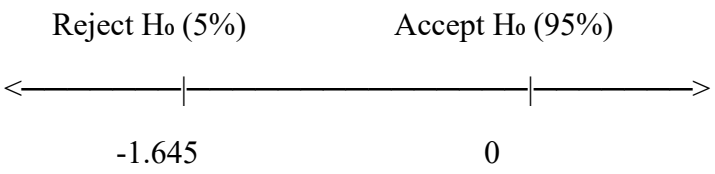
Test Type	Critical Value(s)	Rejection Rule
Left-tailed	Z = −1.645	Reject H0 if z ≤ −1.645
Right-tailed	Z = +1.645	Reject H0 if z ≥ +1.645
Two-tailed	Z = ±1.96	Reject H0 if (z\geq 1.96)

---

2. Graphical Representation

Left-tailed Test (α = 0.05)

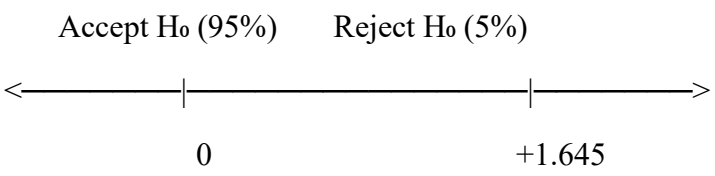
- Reject H0 if the test statistic falls in the **left 5%**.



Right-tailed Test (α = 0.05)

- Reject H0 if the test statistic falls in the **right 5%**.

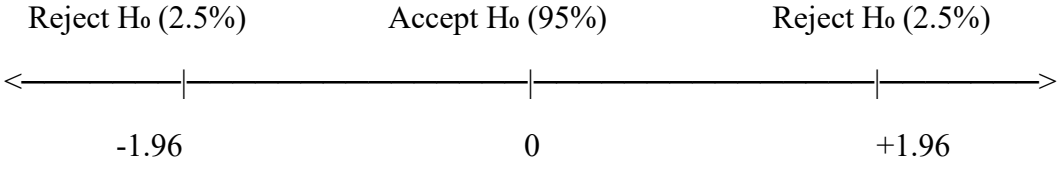
text



Two-tailed Test (α = 0.05)

- Reject Ho if the test statistic is **below -1.96 or above +1.96**.

text



---

3. How to Use the Critical Region in Decision-Making?

1. **Compute the test statistic** (e.g., z-score, t-score).
2. **Compare it to the critical value**:

- If it falls in the **critical region → Reject H0**.
- If it falls in the **acceptance region → Fail to reject H0**.

Example (Right-tailed Z-Test, α = 0.05)

- Suppose z = 2.1
- Critical value = **+1.645**.
- Since **2.1 > 1.645**, we **reject H0**.

---

4. Critical Region vs. P-value

Critical Region Approach	P-value Approach
Compare test statistic to a fixed threshold (critical value).	Compare p-value to α.
Reject H0 if test stat is in the rejection zone.	Reject H0 if p-value ≤ α.
Requires pre-defined α.	Directly gives the probability of observing the result under H0.

---

5. Summary

- The **critical region** is where we reject H0.
- It depends on:
  - **α (significance level)**,
  - **type of test (left, right, two-tailed)**.
- **Decision rule**: Reject H0 if the test statistic falls in the critical region.

3. Types of Hypothesis Tests

Test	When to Use	Example
Z-test	Large sample, known σ	Testing if avg. height = 170cm (σ known)
One - sample t-test	Small sample, unknown σ	Is avg. exam score = 75?
Two-sample t-test	Compare 2 group means	Do men & women earn the same?
Paired t-test	Before vs. after	Did a training program improve scores?
Chi-square	Categorical data	Is voting preference independent of gender?

ANOVA	Compare >2 groups	Do 3 diets lead to different weight loss?

4. Common Errors

- **Type I Error (False Positive):** Rejecting H_0 when it's true (α = probability of this).
- **Type II Error (False Negative):** Failing to reject H_0 when it's false (β).
- **Power (1 – β):** Probability of correctly rejecting H_0.

---

5. Example (t-test in Python)

```
from scipy import stats

# Sample data (before & after treatment)
before = [72, 75, 68, 80, 74]
after = [70, 73, 65, 78, 72]

# Paired t-test
t_stat, p_value = stats.ttest_rel(before, after)

print(f"t-statistic = {t_stat:.2f}, p-value = {p_value:.4f}")
if p_value < 0.05:
    print("Reject H₀: Significant difference after treatment.")
else:
    print("Fail to reject H₀: No significant effect.")
```

Output:

t-statistic = 3.16, p-value = 0.0349

Reject H_0: Significant difference after treatment.

---

6. Key Takeaways

✓ Hypothesis testing evaluates if **observed effects are real** or due to chance.
✓ **Reject H_0 if $p \leq \alpha$** (evidence supports H_1).
✓ Choose the right test based on **data type & sample size**.
✓ Watch out for **Type I/II errors** and statistical power.

Real-World Hypothesis Testing Walkthrough

Scenario: A company claims its new battery lasts **≥100 hours**. A sample of 30 batteries is tested, with an **average of 98 hours** and a **standard deviation of 5 hours**. Is the company's claim valid?

Step 1: Define Hypotheses

- **Null Hypothesis (H_0):**
  "The average battery life is ≥100 hours."
  (Formally: H_0: $\mu \geq 100$)
- **Alternative Hypothesis (H_1):**
  "The average battery life is <100 hours."
  (Formally: H_1: $\mu < 100$) → **One-tailed test**

Step 2: Choose Significance Level

- Let's use **$\alpha = 0.05$** (5% risk of false rejection).

Step 3: Select the Right Test

- **Sample size (n = 30)** → Use a **Z-test** (since $n \geq 30$, CLT applies).
- If $n < 30$, use a **t-test**.

Step 4: Calculate Test Statistic

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{98 - 100}{5/\sqrt{30}} = \frac{-2}{0.912} = -2.19$$

- **Z = -2.19** (negative because sample mean < claimed mean).

Step 5: Find the p-value

- For **Z = -2.19**, the one-tailed p-value = **0.0143** (from Z-table).
- **$p < \alpha$ (0.0143 < 0.05)** → Reject H_0.

Step 6: Make a Decision

- **Reject H_0** → Evidence supports that the **battery life is <100 hours**.
- The company's claim is **not statistically valid**.

Python Verification

```
from scipy import stats

import numpy as np


sample_mean = 98
```

```
pop_mean = 100
sample_std = 5
n = 30

# Calculate Z-score and p-value
z_score = (sample_mean - pop_mean) / (sample_std / np.sqrt(n))
p_value = stats.norm.cdf(z_score)  # One-tailed p-value

print(f"Z-score: {z_score:.2f}, p-value: {p_value:.4f}")

if p_value < 0.05:
    print("Reject H₀: Battery life is significantly <100 hours.")
else:
    print("Fail to reject H₀: No evidence against the claim.")
```

Output:

text

Z-score: -2.19, p-value: 0.0143

Reject H₀: Battery life is significantly <100 hours.

Key Takeaways

1. **H_0 vs. H_1:** Always define *what you're testing*.
2. **One-tailed vs. Two-tailed:**
   - One-tailed (directional: ">", "<") → More power to detect an effect.
   - Two-tailed (non-directional: "≠") → Tests for *any difference*.
3. **Interpretation:**
   - **$p < \alpha$ → Reject H_0** (significant result).
   - **$p \geq \alpha$ → Fail to reject H_0** (no evidence against H_0).

Next Steps:

- Try a **two-sample t-test** (e.g., comparing 2 groups).
- Learn about **effect size** (how large is the difference?).

To determine whether a hypothesis test is **one-tailed (left or right)** or **two-tailed**, you need to examine the **alternative hypothesis (Ha)**. Here's a step-by-step guide:

---

1. Look at the Alternative Hypothesis (Ha)

The **direction** of Ha determines whether the test is **left-tailed, right-tailed, or two-tailed**.

Alternative Hypothesis (Ha)	Type of Test	P-value Calculation
Ha:μ<μ0 (Mean is **less than** a value)	**Left-tailed**	P(Z≤z)
Ha:μ>μ0 (Mean is **greater than** a value)	**Right-tailed**	P(Z≥z)=1−P(Z≤z)
Ha:μ≠μ0 (Mean is **not equal to** a value)	**Two-tailed**	(2 \times P(Z \geq z))

---

2. Examples

Example 1 (Left-tailed Test)

- **Null Hypothesis (H0)**: The average score is **50** (μ=50).
- **Alternative Hypothesis (Ha)**: The average score is **less than 50** (μ<50).
- **Test Type**: **Left-tailed** (only interested in values **below** 50).

Example 2 (Right-tailed Test)

- **Null Hypothesis (H0)**: The average height is **170 cm** (μ=170).
- **Alternative Hypothesis (Ha)**: The average height is **greater than 170 cm** (μ>170).
- **Test Type**: **Right-tailed** (only interested in values **above** 170).

Example 3 (Two-tailed Test)

- **Null Hypothesis (H0)**: The average weight is **60 kg** (μ=60).
- **Alternative Hypothesis (Ha)**: The average weight is **not equal to 60 kg** (μ≠60).
- **Test Type**: **Two-tailed** (interested in values **either above or below** 60).

---

3. Visual Representation

- **Left-tailed**: Rejection region is **only on the left side** of the distribution.
- **Right-tailed**: Rejection region is **only on the right side**.
- **Two-tailed**: Rejection regions are on **both sides**.

https://www.statisticshowto.com/wp-content/uploads/2014/02/tailed-test.png

---

4. How to Decide?

- **If you're testing for "greater than" or "less than" → One-tailed (right or left).**
- **If you're testing for "not equal to" → Two-tailed.**

Final Summary

If Ha says...	Test Type	P-value Formula
μ<μ0	Left-tailed	P(Z≤z)
μ>μ0	Right-tailed	1−P(Z≤z)
μ≠μ0	Two-tailed	(2 \times P(Z \geq | z |))

Key Takeaway

- **Check Ha** to determine if the test is **left, right, or two-tailed**.
- **One-tailed tests** are directional (only one side matters).
- **Two-tailed tests** check for **any difference** (both sides matter).

1. Two-Sample Independent T-test

Scenario:
A researcher wants to compare the mean test scores of two independent groups:

- **Group A** (n_1 = 20, mean = 78, SD = 5)
- **Group B** (n_2 = 25, mean = 82, SD = 6)

Hypotheses:

- H0: μ1=μ2 (No difference in means)
- H1: μ1≠μ2 (Two-tailed test)

Solution:

1. **Calculate pooled variance (assuming equal variances)**:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(19)(5^2) + (24)(6^2)}{43} = \frac{475 + 864}{43} = 31.14$$

2. **Compute the t-statistic**:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{78 - 82}{\sqrt{31.14\left(\frac{1}{20} + \frac{1}{25}\right)}} = \frac{-4}{1.77} = -2.26$$

3. **Find critical t-value (df = 43, α = 0.05)**:
   - From t-table, $t_{crit} \approx \pm 2.02$ (two-tailed).
4. **Conclusion**:
   - Since |t|=2.26 > 2.02, **reject H0**.

- The means are significantly different.

WE ARE STILL WORKING ON BELOW 2 EXAMPLES

2. Two-Sample Related (Paired) T-test

Scenario:
A fitness trainer measures the weight (kg) of 10 participants before and after a program:

Participant	Before	After	Difference (d)
1	70	68	-2
2	75	72	-3
...	...	...	...
10	80	78	-2

Hypotheses:

- H0: μd = 0 (No difference)
- H1: μd ≠ 0 (Two-tailed)

Solution:

1. **Calculate mean difference ($\bar{d}$) and SD of differences**:
   - Suppose $\bar{d} = -2.5$, sd = 1.2
2. **Compute t-statistic**:

T = $\bar{d}$s d/n = −2.51.2/10 = −2.50.38 =−6.58

3. **Find critical t-value (df = 9, α = 0.05)**:
   - From t-table, tcrit = ±2.262.
4. **Conclusion**:
   - Since |t|=6.58 > 2.262 , **reject H0**.
   - The program significantly reduced weight.

---

3. One-Way ANOVA Test

Scenario:
A study compares the effect of three diets (A, B, C) on weight loss (kg):

Diet A	Diet B	Diet C
3	5	6
4	6	7
2	4	5
...	...	...

Hypotheses:

- H0: $\mu A = \mu B = \mu C$ (All means equal)
- H1: At least one mean differs

Solution:

1. **Calculate group means and overall mean**:
    - Suppose:
        - $\bar{X}A=3.0$, $\bar{X}B=5.0$, $\bar{X}C=6.0$
        - Overall mean ($\bar{X}$total) = 4.67
2. **Compute Sum of Squares (SS)**:
    - **Between-group (SSB)**:

$SSB=n\sum(\bar{X}i-\bar{X}total)2=10[(3-4.67)2+(5-4.67)2+(6-4.67)2]=46.67$

    - **Within-group (SSW)**:

$SSW=\sum(Xij-\bar{X}i)2=20$

3. **Compute Mean Squares (MS)**:
    - MSB = SSBk−1 = 46.67/2 =23.34
    - MSW = SSWN−k = 20/27 =0.74
4. **Compute F-statistic**:

F= MSBMSW =23.340.74=31.54

5. **Compare to critical F-value ($df_1 = 2$, $df_2 = 27$, $\alpha = 0.05$)**:
    - From F-table, Fcrit≈3.35.
    - Since F=31.54>3.35, **reject H0**.
6. **Conclusion**:
    - At least one diet leads to significantly different weight loss.

Summary Table

Test Type	Hypotheses Tested	Key Formula	Decision Rule		
Independent T-test	$\mu 1=\mu 2 \mu 1=\mu 2$	$t=\bar{X}1-\bar{X}2sp1n1+1n2 t=spn11+n21$ $\bar{X}1-\bar{X}2$	Reject H0$H0$ if (	t	> t_{crit})
Paired T-test	$\mu d=0$	$t=\bar{d}sd/n$	Reject H0$H0$ if (	t	> t_{crit})
One-Way ANOVA	$\mu 1=\mu 2=\mu 3$	$F=MSBMSW$	Reject H0 if F>Fcrit		

Applications of Hypothesis Testing

Hypothesis testing is a fundamental statistical method used to make **data-driven decisions** in various fields. It helps determine whether observed effects are **real (statistically significant)** or due to **random chance**. Below are key applications across different domains:

1. Medicine & Healthcare

- **Drug Efficacy Testing**
    - *Example*: Testing if a new drug lowers blood pressure more than a placebo.
    - H0: Drug effect = Placebo effect
      Ha: Drug effect > Placebo effect (Right-tailed test)
- **Disease Diagnosis**
    - *Example*: Does a new diagnostic tool detect cancer more accurately than the standard method?

2. Business & Economics

- **A/B Testing (Marketing & Web Design)**
    - *Example*: Does a **new webpage design** increase conversion rates compared to the old one?
    - H0: New design conversion rate = Old design
      Ha: New design conversion rate > Old design (Right-tailed test)
- **Product Quality Control**
    - *Example*: A factory tests if the **mean weight of cereal boxes** meets the advertised 500g.
    - H0: $\mu = 500g$
      Ha: $\mu \neq 500g$ (Two-tailed test)

3. Social Sciences & Psychology

- **Survey Analysis**
    - *Example*: Is there a **gender difference** in job satisfaction scores?

- H0: Mean satisfaction (Men) = Mean satisfaction (Women)
  Ha: Means are not equal (Two-tailed test)

- **Educational Research**
  - *Example*: Does a **new teaching method** improve student test scores?

4. Engineering & Manufacturing

- **Process Improvement**
  - *Example*: Does a **new manufacturing process** reduce defect rates?
  - H0*H*0: Defect rate (New) = Defect rate (Old)
    Ha*Ha*: Defect rate (New) < Defect rate (Old) (Left-tailed test)
- **Reliability Testing**
  - *Example*: Does a new engine design last longer than the previous model?

5. Finance & Risk Management

- **Stock Market Analysis**
  - *Example*: Does a **new trading strategy** yield higher returns than the market average?
  - H0*H*0: Strategy return = Market return
    Ha*Ha*: Strategy return > Market return (Right-tailed test)
- **Credit Risk Assessment**
  - *Example*: Do borrowers with **high credit scores** default less often?

6. Environmental Science

- **Climate Change Studies**
  - *Example*: Has **average global temperature** increased over the past century?
  - H0*H*0: No change in temperature
    Ha*Ha*: Temperature has increased (Right-tailed test)
- **Pollution Control**
  - *Example*: Does a **new filter** reduce factory emissions below regulatory limits?

7. Public Policy & Government

- **Policy Impact Evaluation**
  - *Example*: Did a **new traffic law** reduce accident rates?
  - H0*H*0: Accident rate (After) = Accident rate (Before)
    Ha*Ha*: Accident rate (After) < Accident rate (Before) (Left-tailed test)
- **Census & Population Studies**
  - *Example*: Is there a **significant difference** in income levels between two regions?

Key Takeaways

✅ **Hypothesis testing is used to:**

- Validate assumptions (e.g., "This drug works").
- Compare groups (e.g., "Does Method A perform better than Method B?").
- Detect changes (e.g., "Has customer satisfaction improved?").

✅ **Common Tests Used:**

- **Z-test / T-test** (Means)
- **Chi-square test** (Categorical data)
- **ANOVA** (Comparing multiple groups)

STATISTICS QUESTIONNAIRE

Theory Questionnaire on Statistics for Data Analytics

Section A: Basic Concepts

1. Define **Statistics** and explain its importance in Data Analytics.
2. Differentiate between **Descriptive Statistics** and **Inferential Statistics**.
3. What are the different **types of data**? Explain with examples.
4. Define **Population** and **Sample**. Why is sampling important in data analytics?
5. What are **variables** in statistics? Classify them with examples.

Section B: Descriptive Statistics

6. Explain the measures of **Central Tendency** (Mean, Median, Mode) and their significance.
7. What are the measures of **Dispersion**? Explain Variance and Standard Deviation.
8. How do **Skewness** and **Kurtosis** help in understanding data distribution?
9. What is a **Box Plot**? How does it help in identifying outliers?
10. Explain the concept of **Correlation** and its coefficient.

Section C: Probability & Distributions

11. Define **Probability**. How is it used in predictive analytics?
12. Explain **Bayes' Theorem** with an example.
13. Differentiate between **Discrete** and **Continuous** probability distributions.
14. Explain the properties and applications of **Normal Distribution**.
15. What is **Central Limit Theorem (CLT)**? Why is it important in statistics?

Section D: Inferential Statistics

16. Define **Hypothesis Testing**. Explain **Null (H_0)** and **Alternative (H_1)** hypotheses.
17. What are **p-values** and **significance levels (α)**? How are they used in hypothesis testing?
18. Explain **Confidence Intervals** and their interpretation.
19. Differentiate between **Z-test** and **T-test**. When is each used?
20. What is **ANOVA**? Explain its application in data analytics.